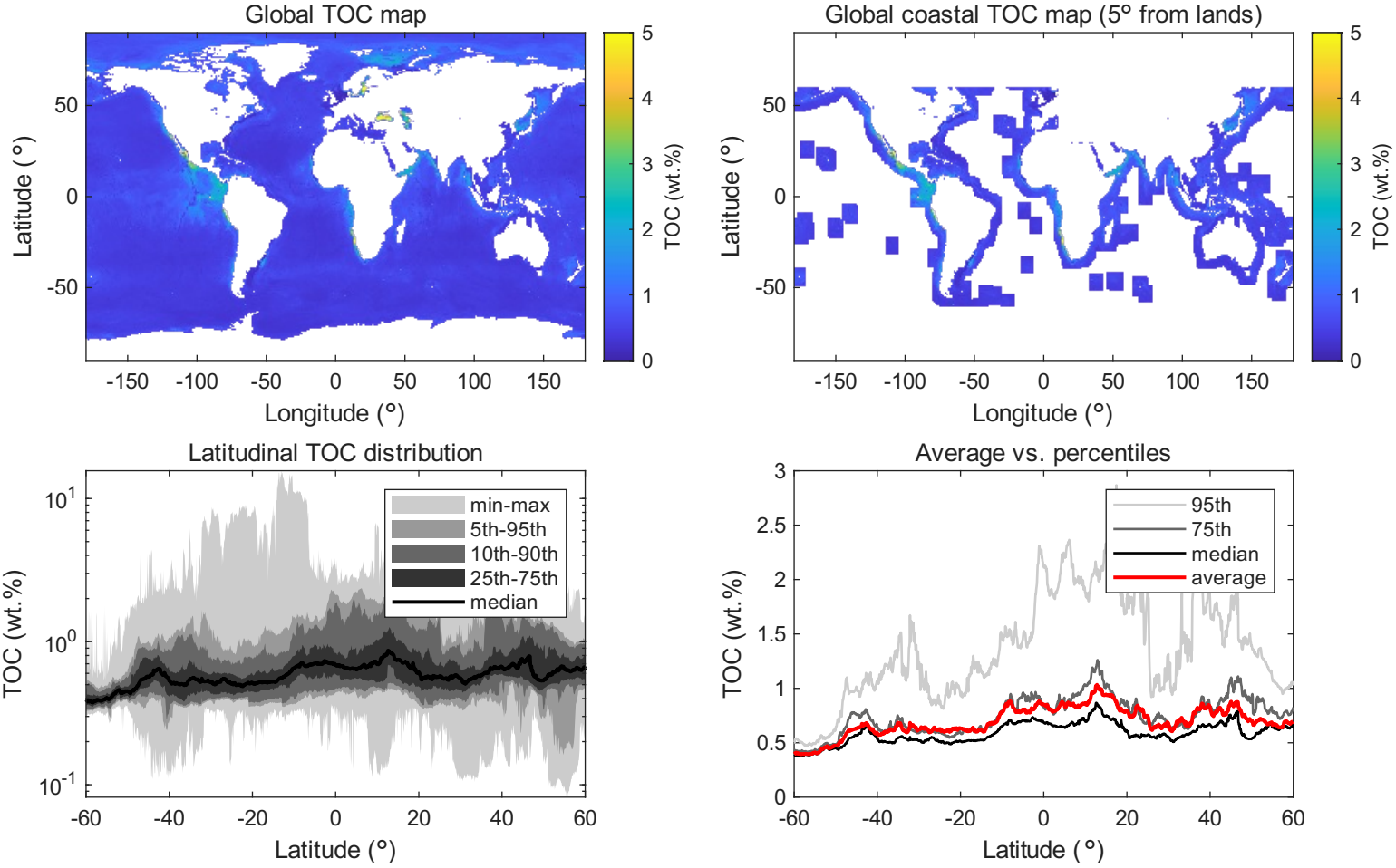
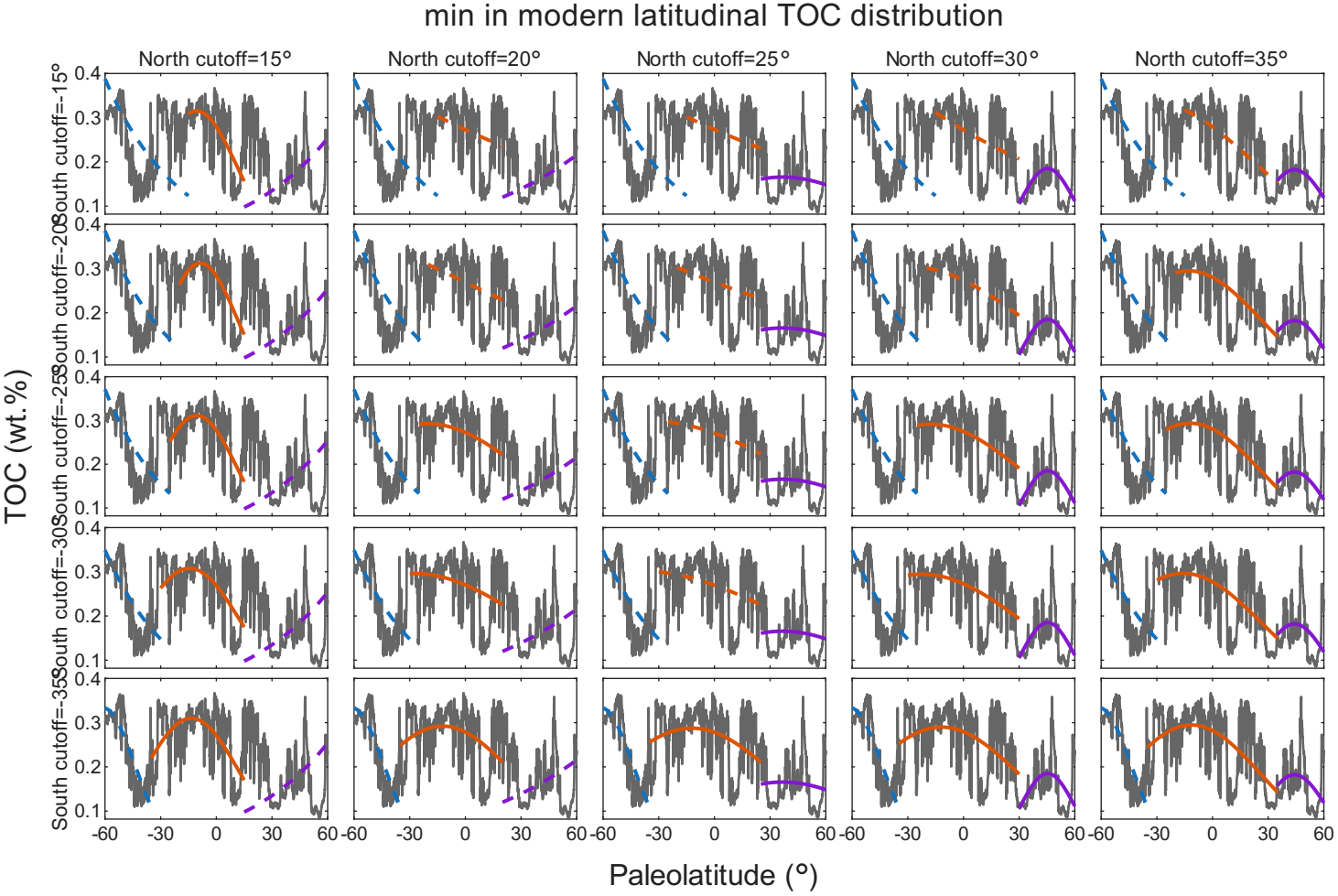
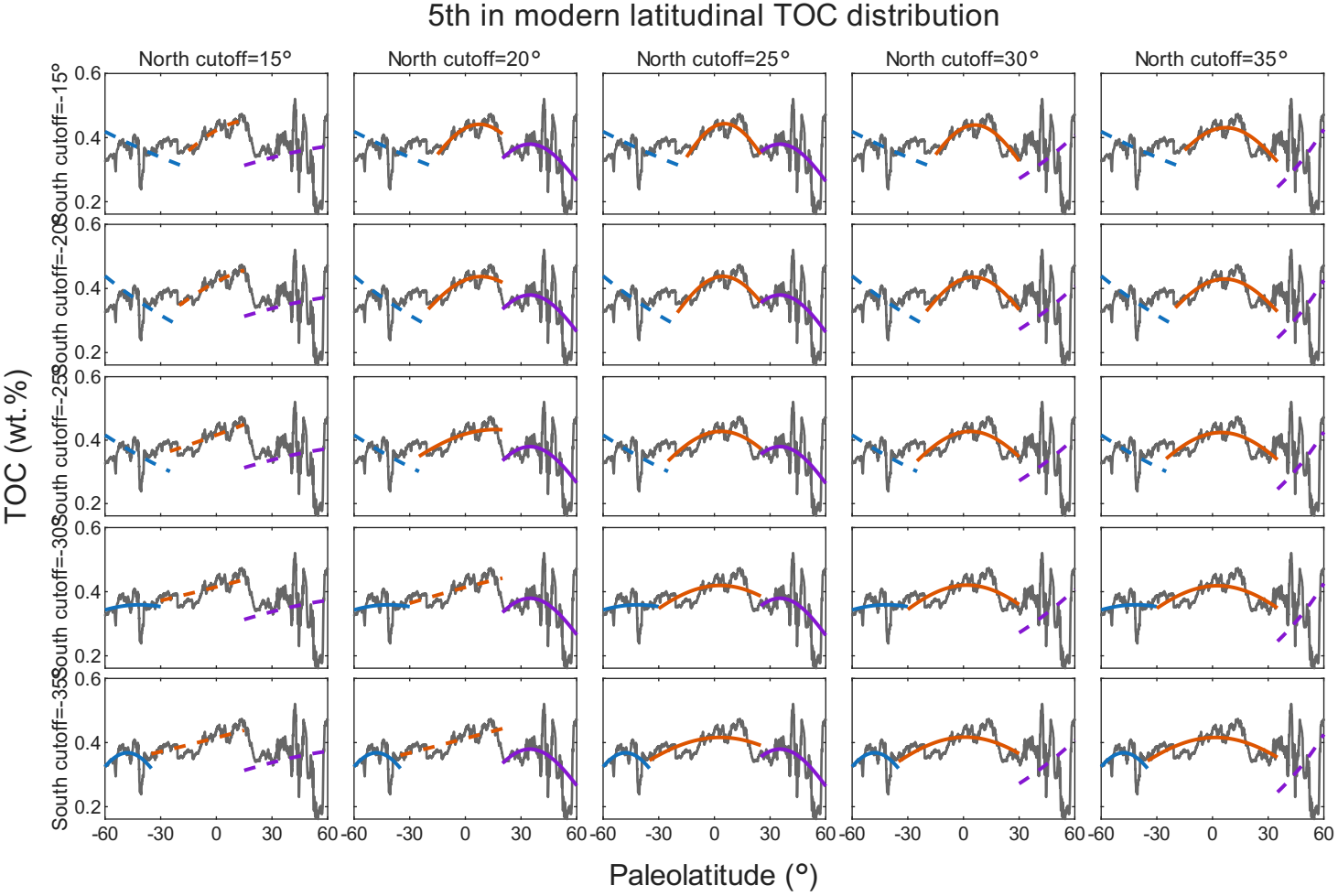
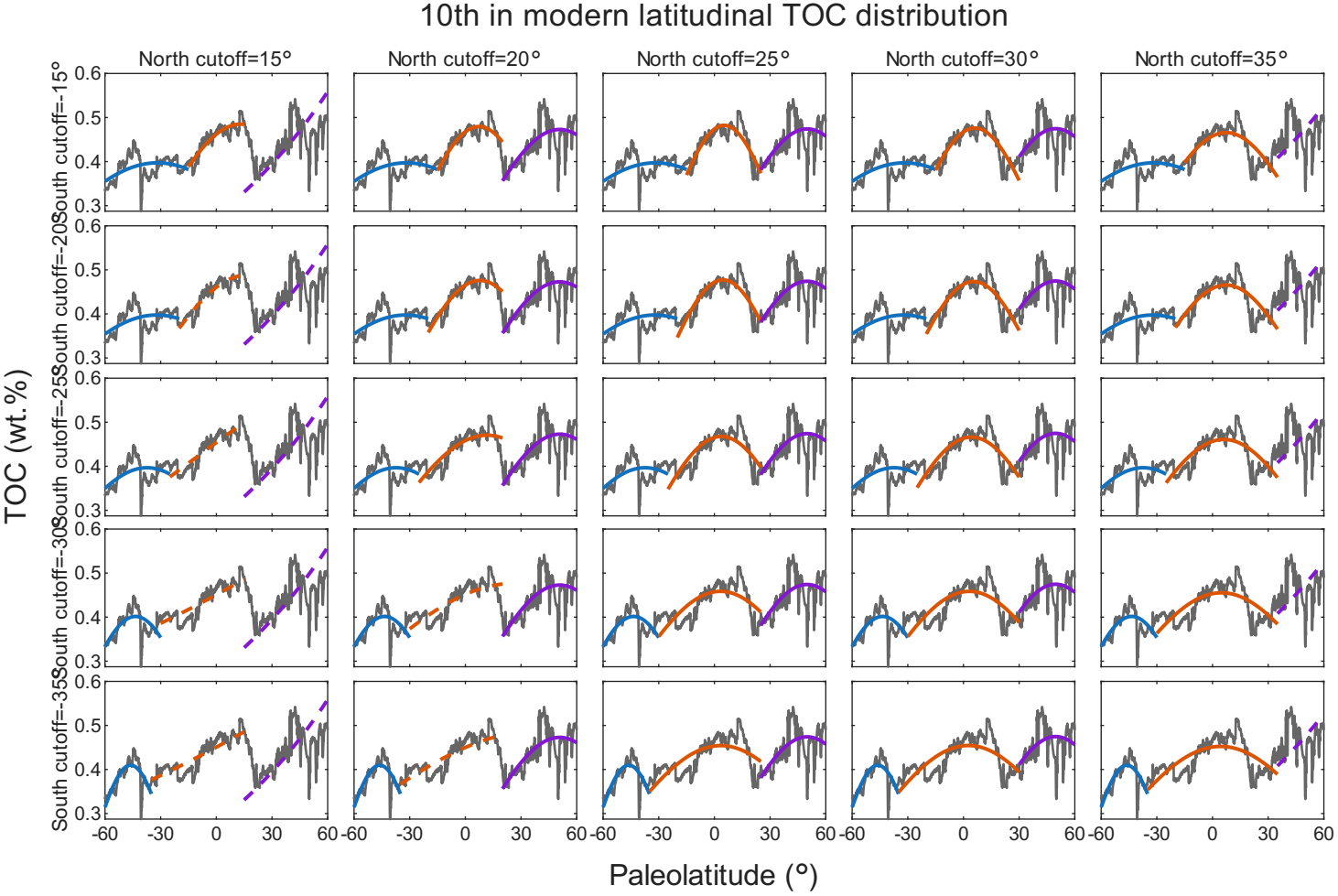
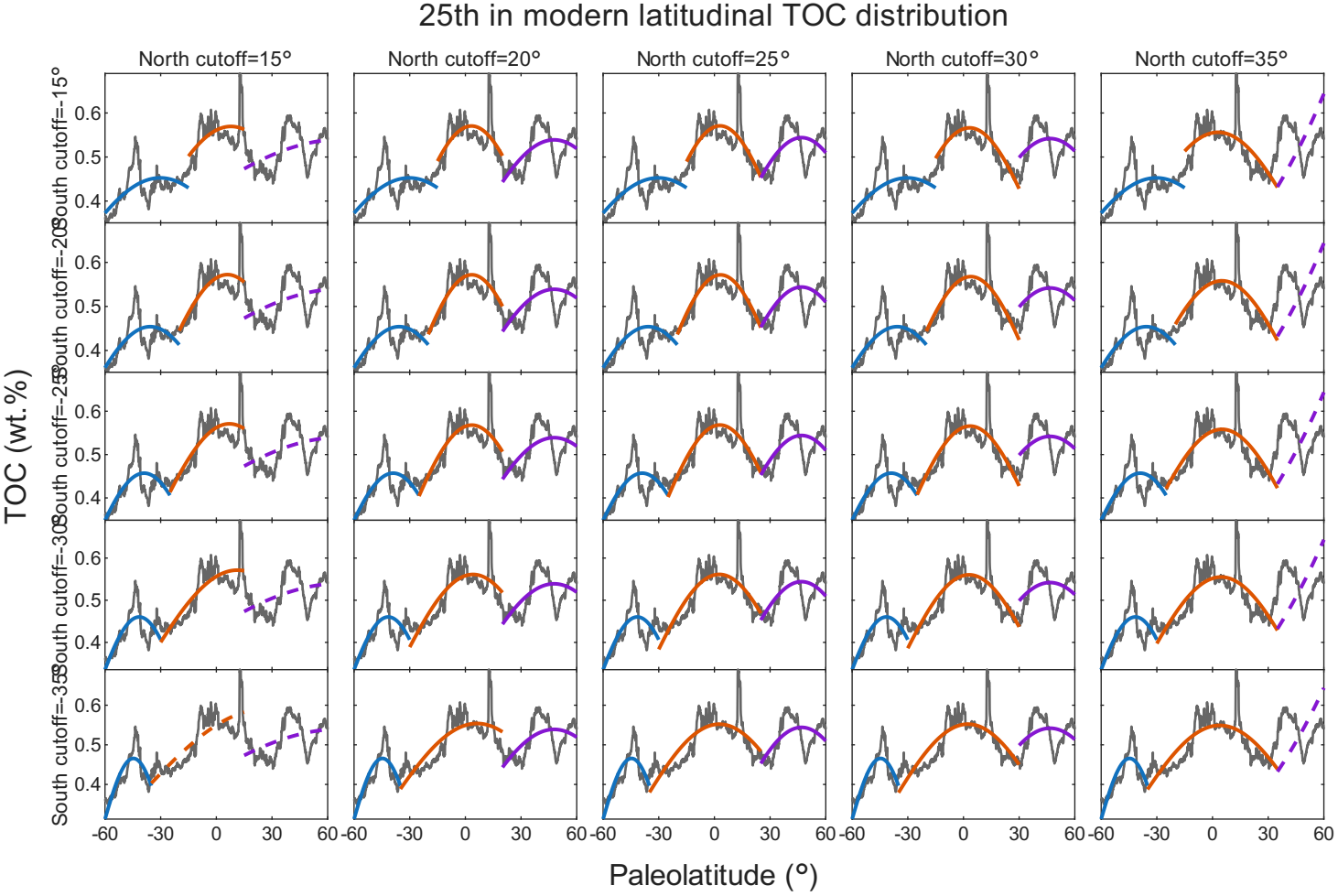
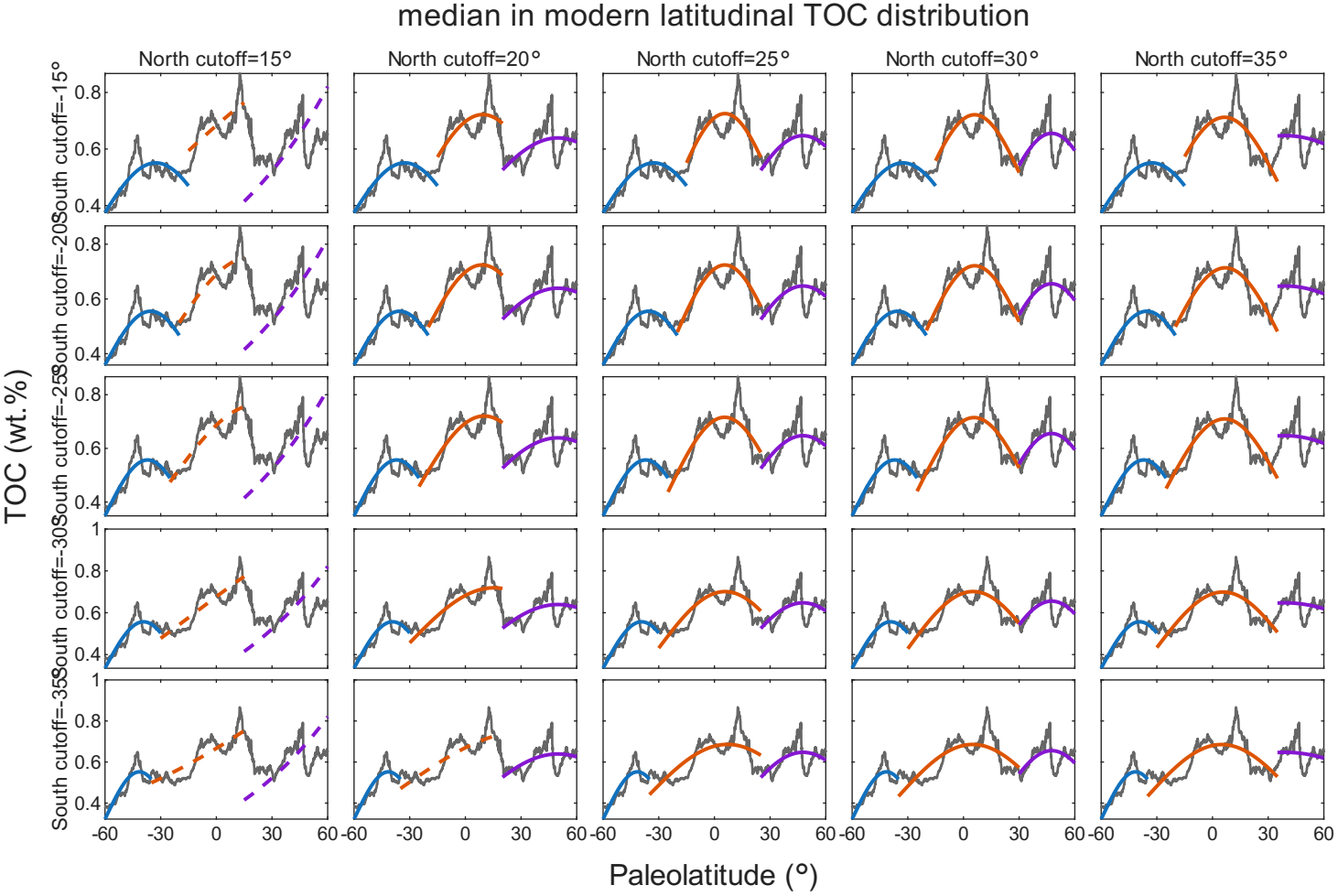

Global TOC map
Latitude (°)
Longitude (°)
TOC (wt.%)
Global coastal TOC map (5° from lands)
Latitudinal TOC distribution
min-max
5th-95th
10th-90th
25th-75th
median
Average vs. percentiles
95th
75th
median
average

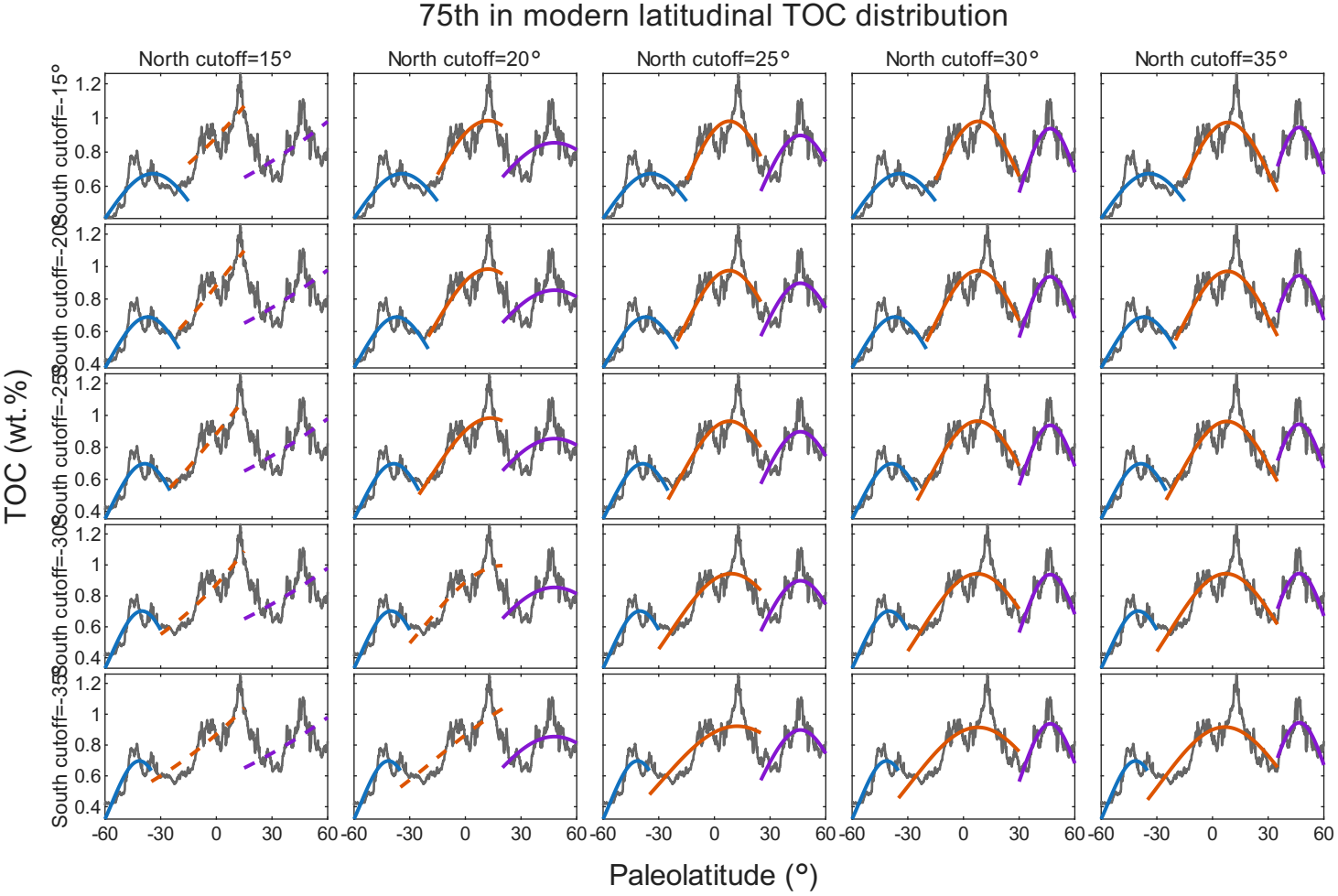

min in modern latitudinal TOC distribution
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
0.4
0.3
0.2
0.1
-60
-30
0
30
60
Paleolatitude (°)

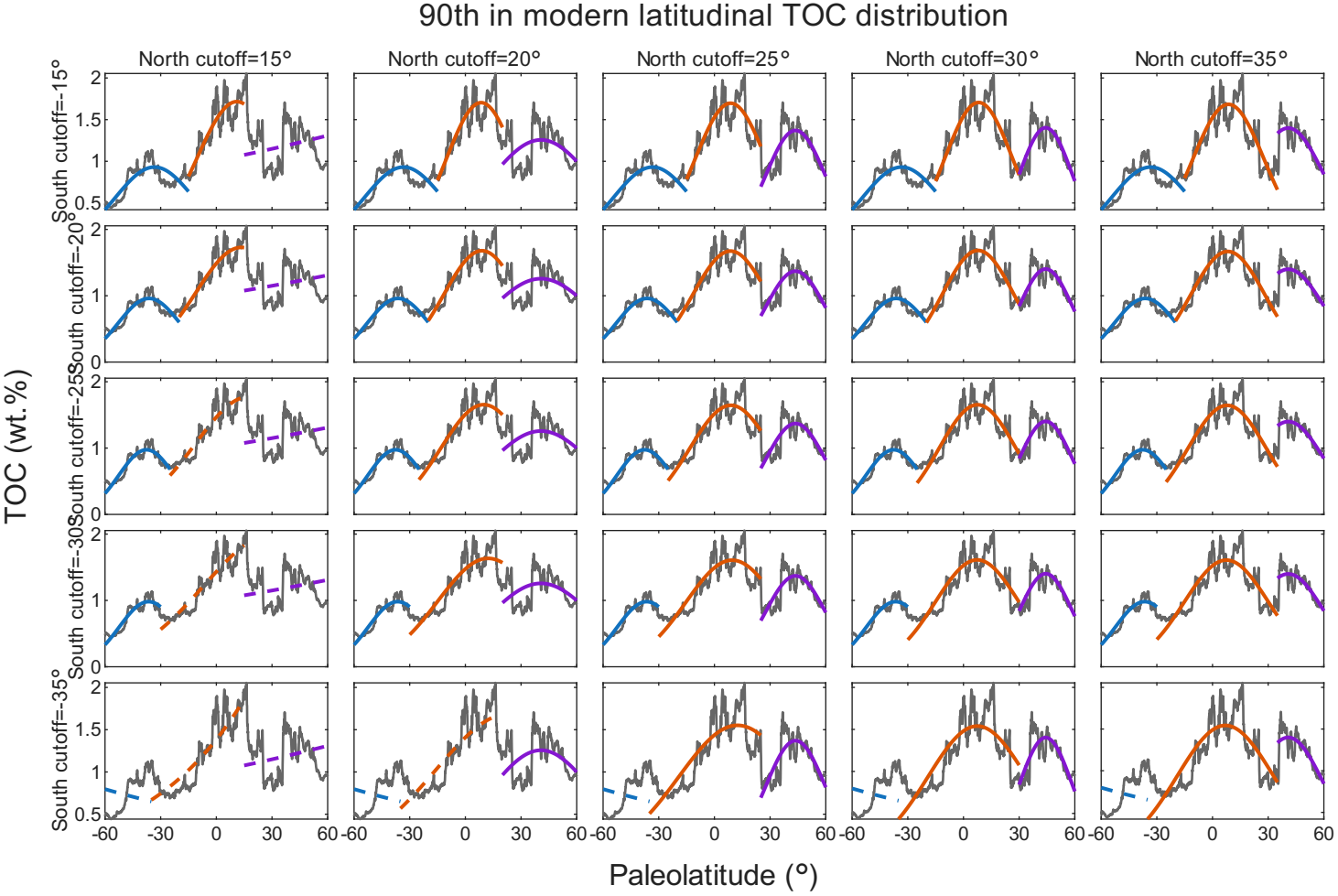

5th in modern latitudinal TOC distribution
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
0.6
0.4
0.2
-60
-30
0
30
60
Paleolatitude (°)

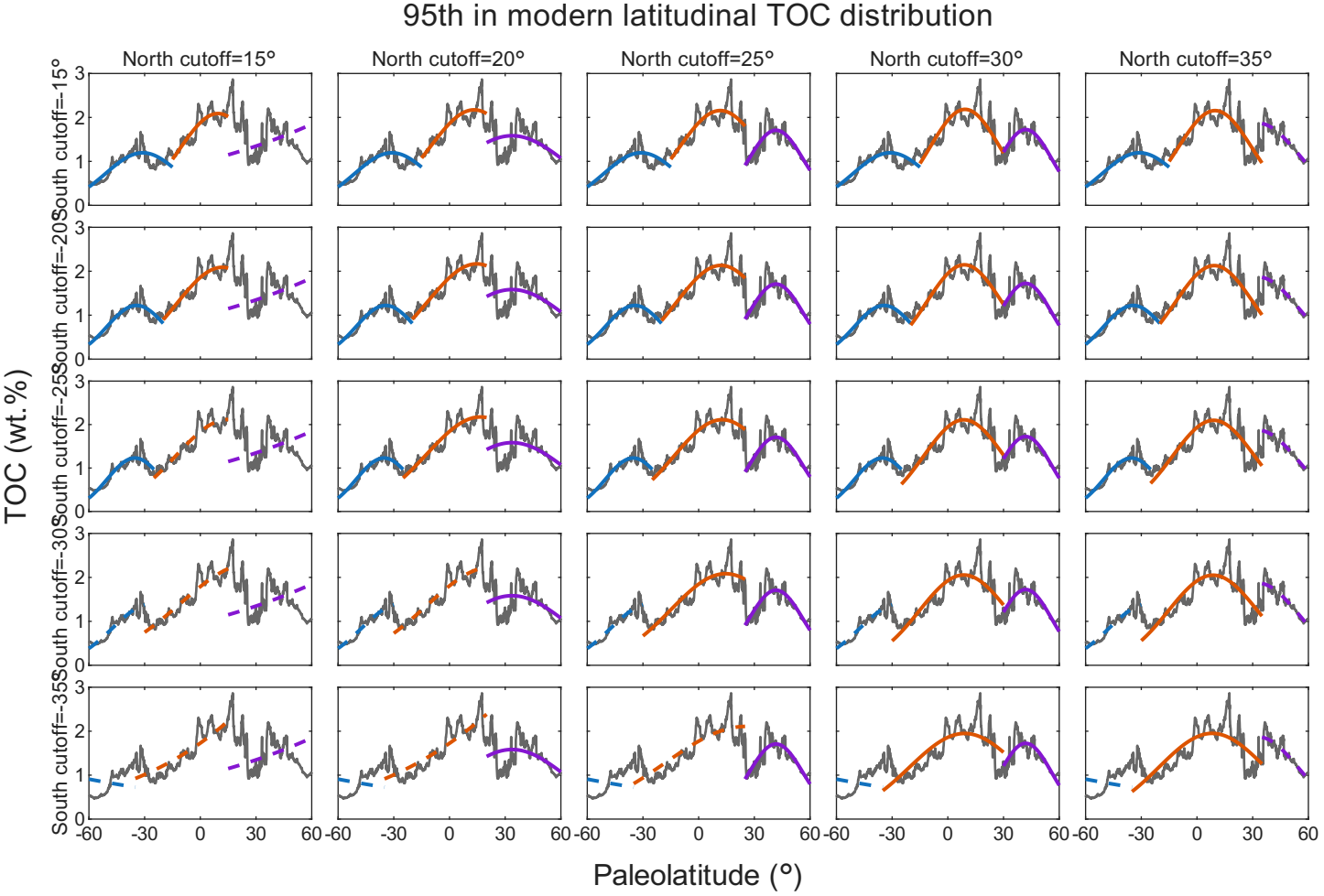

10th in modern latitudinal TOC distribution
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
0.3
0.4
0.5
0.6
-60
-30
0
30
60
Paleolatitude (°)

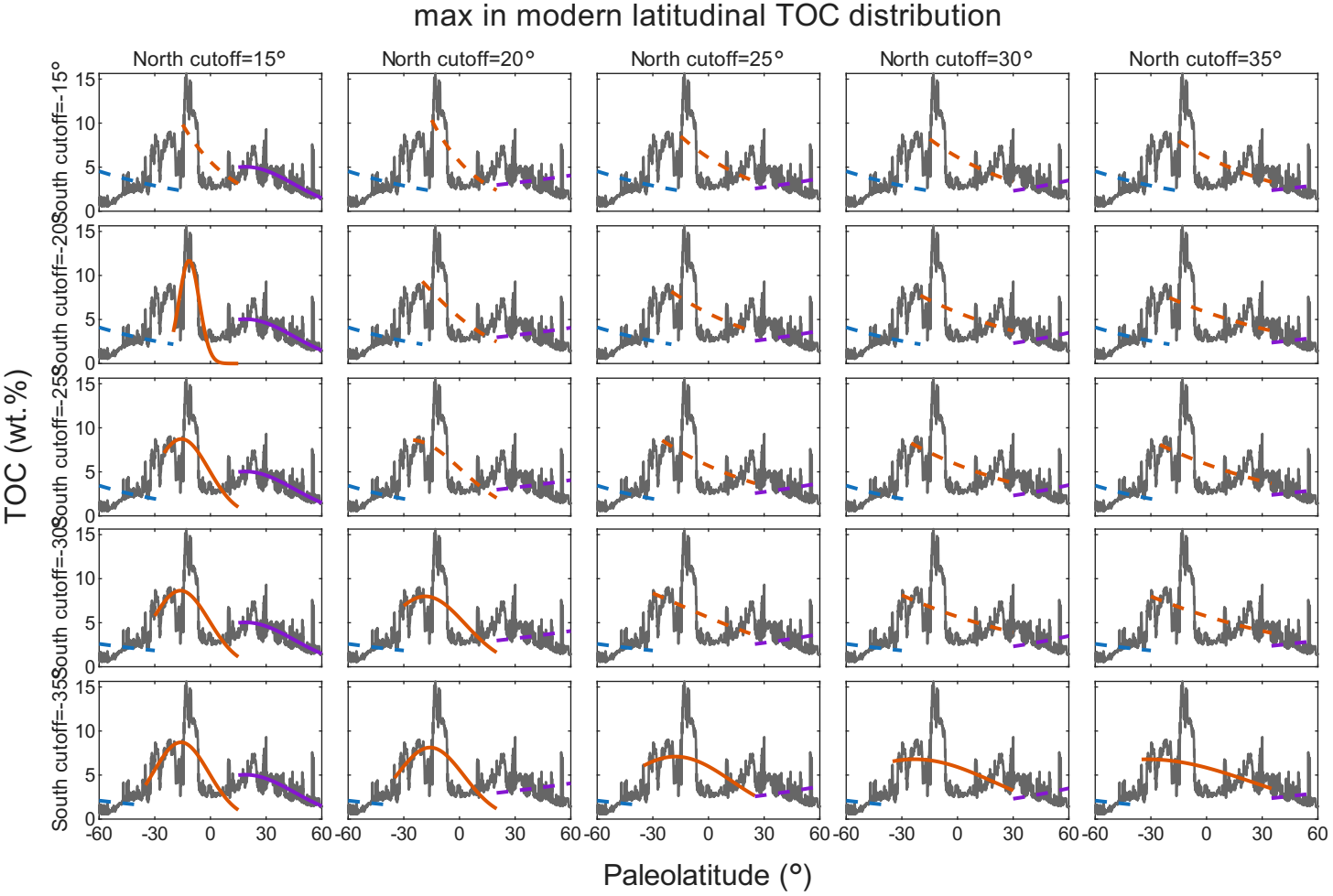

25th in modern latitudinal TOC distribution
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
0.4
0.5
0.6
-60
-30
0
30
60
Paleolatitude (°)

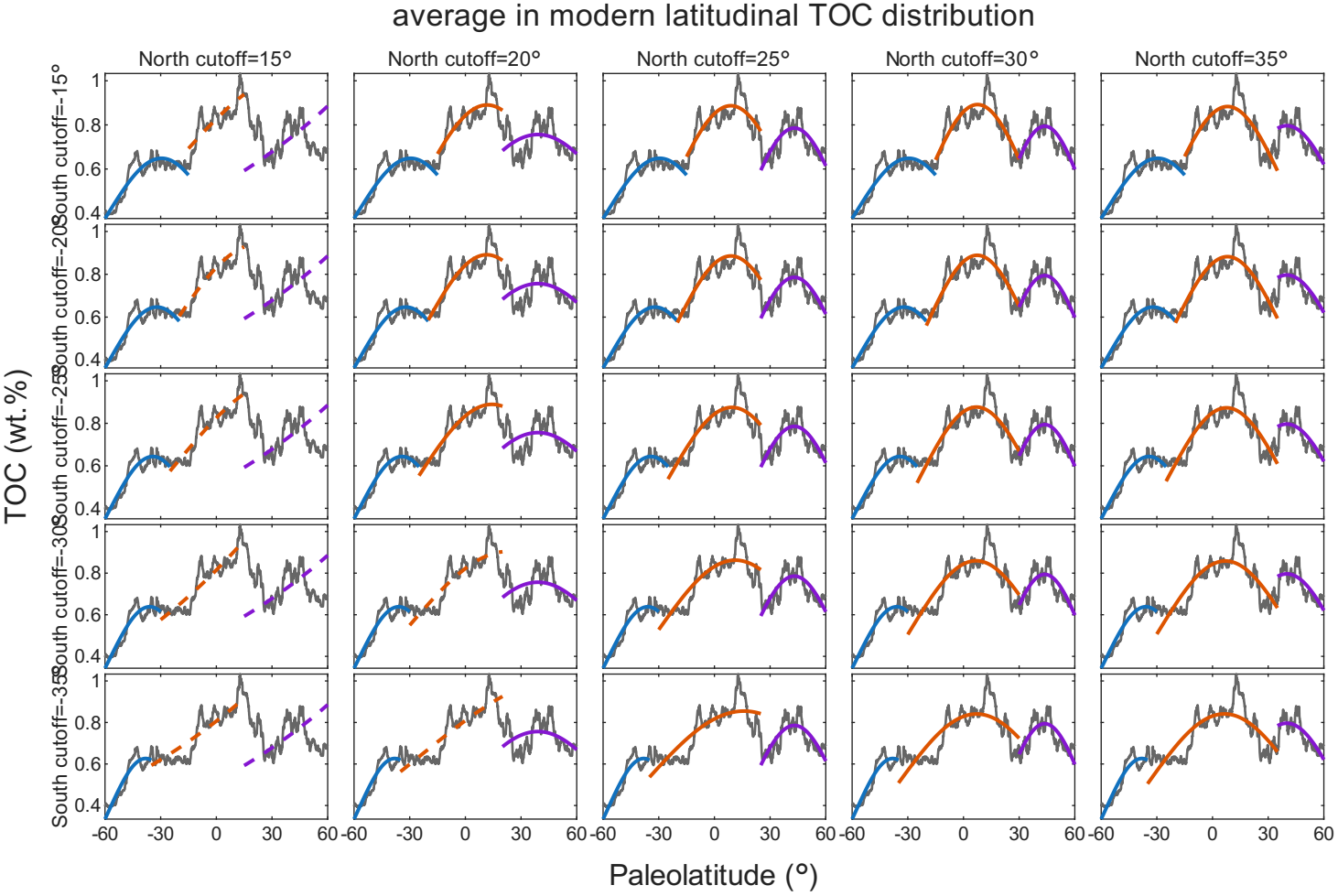

median in modern latitudinal TOC distribution
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)

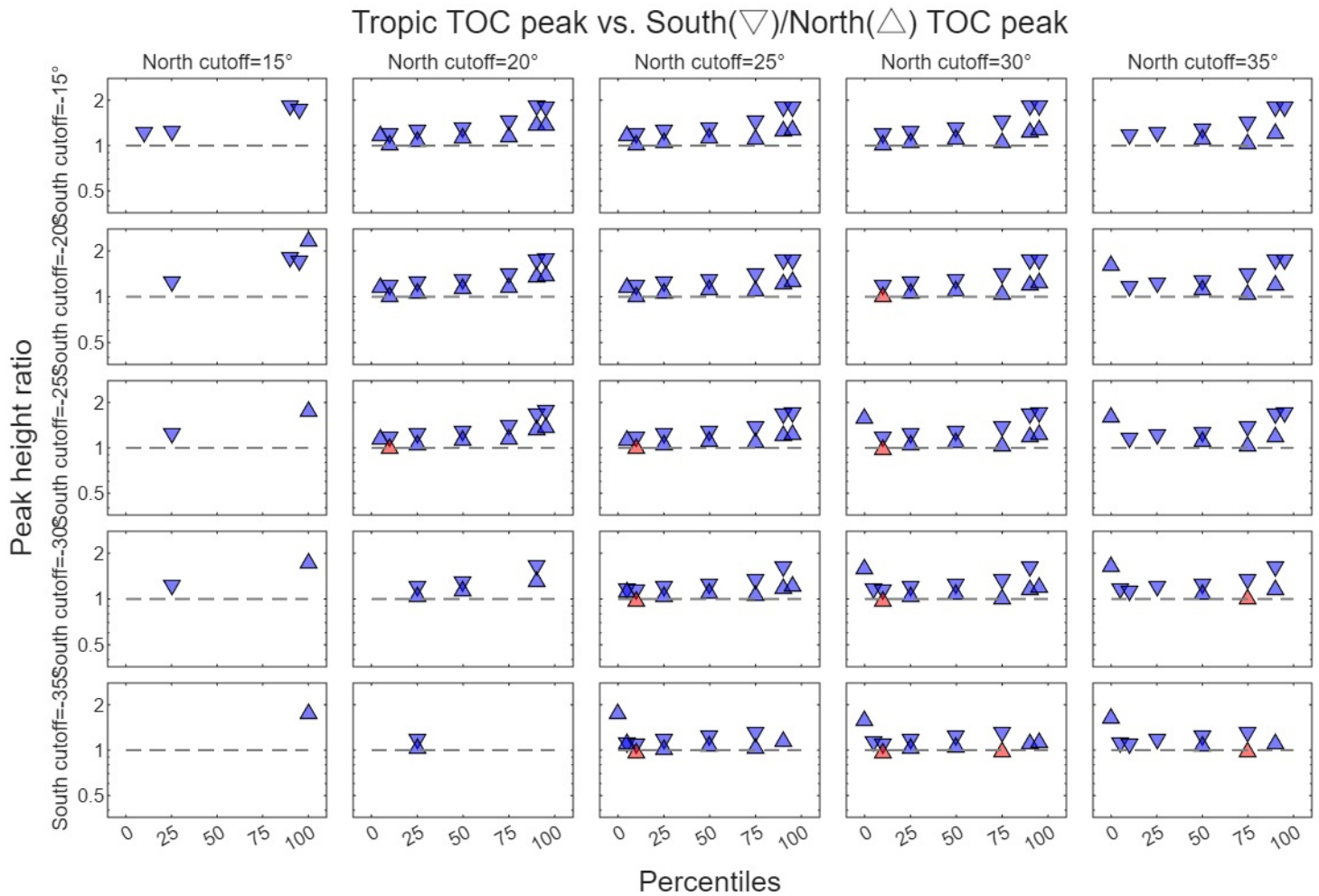

75th in modern latitudinal TOC distribution
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)


90th in modern latitudinal TOC distribution
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
-60
-30
0
30
60
0.5
1
1.5
2


95th in modern latitudinal TOC distribution
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
0
1
2
3
-60
-30
0
30
60


max in modern latitudinal TOC distribution
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
0
5
10
15
-60
-30
0
30
60


average in modern latitudinal TOC distribution
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
1
0.8
0.6
0.4
-60
-30
0
30
60


Tropic TOC peak vs. South(▽)/North(△) TOC peak
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
Peak height ratio
0.5
1
2
0
25
50
75
100
Percentiles